

2014

M B A

4th Semester Examination

**Subject : MARKETING RESEARCH AND
FORECASTING TECHNIQUES**

(Specialization : Marketing Management)

PAPER—M—402

Full Marks : 100

Time : 3 Hours

The figures in the right-hand margin indicate full marks.

*Candidates are required to give their answers in their
own words as far as practicable.*

Illustrate the answers wherever necessary.

Write the answers to Questions of each Half in separate books.

(First Half)

(Marks : 50)

1. Answer any four of the followings : 4×5

- (a) Why is it important to define the marketing research problem appropriately ?**

(Turn Over)

- (b) What are the differences between research questions and hypothesis ?
- (c) "Researcher should obtain secondary data before primary data" — Justify.
- (d) An organisation has a total of 900 employees in 5 different categories. The no of employees of each categories are :
- (a) Manager — 90
 - (b) Assistant Manager — 100
 - (c) Supervisor — 140
 - (d) Staff — 120
 - (e) Workers — 450

How many from each category should be included in a stratified random sample of size 90 ?

- (e) Draw a random sample of size 10 (with replacement) from the following data :

45	24	43	17	5	28	27	21
33	26	24	14	34	21	25	48
26	27	35	8	30	26	30	28
20	13	23	36	38	20	25	31
11		46					
35		38					
21		27					
24		18					

You may use the random sampling numbers given below :

5905 3364 2689 2453 4826 3508
3042 7130 5370 4683 5872 1143

Calculate the sample mean and population mean.

- (f) From the following information calculate the response rate of a survey.

The sample consisted of 2,000 telephone numbers that were generated randomly,

Number of completed interviews = 750

Number of eligible respondents = 1100

Number of ineligible respondents = 500

Not ascertained (NA) = 400

2. Answer any *two* of the following : 2×10

- (a) How is ordinal scale different from interval scale ?
Design a questionnaire to measure the effectiveness of advertising by using Likert rating and Semantic differential scale. 4+6
- (b) consider the following 5 brands of bath soap; Dove, Camay, Lux, Ivory and Pamolive. Preference of 400 respondents has plotted in the following table. Calculate the most Preferred brand of bath soap. (Use Z table to calculate the value)

	Dove	Camay	Lux	Ivory	Pamolive
Dove	0	240	320	180	380
Camay	160	0	190	250	230
Lux	80	210	0	150	190
Ivory	220	150	250	0	205
Pamolive	20	170	210	195	0

- (c) To determine the effectiveness of the advertising campaign for a new product, management would like to know what percentage of the households are aware of the new product. The ad agency thinks that this figure is as high as 70 percent. The management would like a 97 percent confidence interval and a margin of error not greater than $\pm 2\%$.

- (i) What sample size should be used for this study?
- (ii) If the management wants to be 99% confidence level how will the sample size change?

6+4

[Internal Assessment : 10 Marks]

(Second Half)**(Marks : 50)****3. Answer any four questions :****4×5**

- (a) What is marketing decision support system ?
- (b) Give a hypothetical example of cluster analysis in marketing research.
- (c) Gives the following coefficients $r_{12} = .41$, $r_{23} = .71$, $r_{13} = .5$. Find the value of $r_{12.3}$ and interpret the result.
- (d) Write a short note on territory based sales analysis in marketing research.
- (e) A random sample of 100 firms was taken to find out the average sale per customer. The sample mean was found to be Rs. 300 and the standard deviation Rs. 100. Construct an interval estimate of the population mean with a confidence level of 95.44 percent.
- (f) Two salesmen A and B are employed by a company. The comparative data pertaining to sales made by the two salesmen are as follows :

	<i>Salesman A</i>	<i>Salesman B</i>
No of sales	30	35
Average sales (Rs.)	600	700
Standard deviation (Rs.)	50	40

Do the average sales of the two salesmen differ significantly ?

4. Answer any *two* questions : 2×10

(a) From the following data, fit a multiple regression equation :

<i>Sales territory</i>	<i>Sales (lakh Rs.)</i>	<i>Advertising ('000 Rs.)</i>	<i>No of selling agents</i>
1	120	50	15
2	150	60	20
3	90	40	12
4	70	20	8
5	130	60	14

(b) A random sample of five motor car tyres is taken from each of a brands manufactured by three companies the life time of these tyres is given below on the basis

of the data, test whether the average lifetime of the 3 brands of tyres are equal or not.

Lifetime of Tyres ('000 miles)

<i>Brand</i>	<i>A</i>	<i>B</i>	<i>R</i>
	35	32	33
	34	31	33
	34	31	32
	33	28	32
	34	29	33

(Given $F_{.05} = 3.89$ and $F_{.01} = 6.93$)

- (c) A random sample of 500 students were classified according to economic condition of their family and also according to merit, as shown below :

<i>Merit</i>	<i>Economic Condition</i>			<i>Total</i>
	<i>Rich</i>	<i>Middle class</i>	<i>Poor</i>	
Meritorious	43	137	61	240
Not meritorious	58	113	89	260
Total	100	250	150	500

Test whether the two attributes Merit and Economic Condition are associated or not

(Given, $f_{.05}^2 = 5.99$ o $f_{.01}^2 = 9.21$ for $2df$)

[Internal Assessment : 10 Marks]
